

CENTER FOR RESEARCH AND DEVELOPMENT

CRAD Module

End-to-End Process Assessment & Capability Checklist

Institution: Bestlink College of the Philippines

System: SMS 2 Capstone (Student Management System)

Module path: /modules/crad/

Database: crad_db (MariaDB) + shared sms2_db users/roles

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Prepared as: Professional web developer process assessment

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1. What CRAD Is

CRAD (Center for Research and Development) is the SMS 2 subsystem that manages institutional research operations for Bestlink College. It is not a single CRUD page ? it is a multi-stage workflow engine spanning student submissions, faculty advising, panel defense, manuscript approval, grants/funding, publications/IP, analytics, and notifications.

Two parallel tracks

- ? Capstone / title research: CRAD FORM S2 V3 title approval ? official group registry ? progress/defense ? manuscript/publication.
- ? Research grants: publish calls ? applications ? evaluation ? 6-step approval ? funding ? milestones ? outputs/IP.

Primary business goals

- ? Triple digital-signature title approval (Adviser ? Research Coordinator ? CRAD Officer) on title_approvals.
- ? Register research groups (RG-YYYY-NNN) and TAP/proposal numbers after full approval.
- ? Assign coordinators, advisers, and panel members; notify stakeholders.
- ? Monitor chapters, milestones, progress updates, and AI-assisted document review.
- ? Schedule pre-oral and final defenses; capture evaluations and revision cycles.
- ? Approve final manuscripts and archive outputs in repository / publications.
- ? Run the grant funding pipeline with sequential institutional sign-off.

2. Actors & Roles

Access is enforced through authentication.php + role_permissions (RBAC). Key roles:

Student

Title form (Student Portal), chapters, progress, final manuscript; apply for grants.

Adviser

First title sign-off; progress review; AI analysis; grant approval step 1.

Research Coordinator

Second title screening; adviser/panel assignment (restricted CRAD nav).

CRAD Officer

Final title registry sign-off; groups; defense; grants; analytics; repository.

Panel / Grammarian / Research Director

Chapter & defense evaluation; faculty monitoring.

review_committee / research_grant

Grant scoring and grant-office management views.

Dept. Chair / Dean(hr) / Research Office / VPAA / Finance

Grant approval chain (steps 2?6).

Super Admin

User/RBAC, security settings, backups, audit oversight.

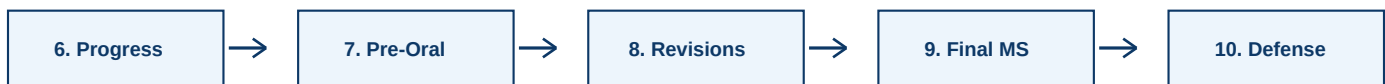
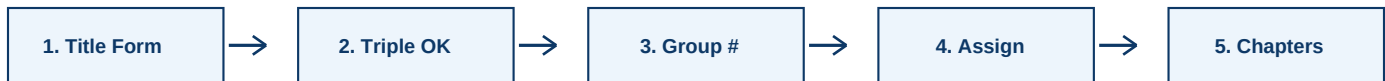
3. End-to-End Research Lifecycle

Primary modern path uses title_approvals (triple digital signature). Legacy research_proposals (CRD-YYYY-NNNNN) still exists alongside it.

Title approval chain (official registry gate)



Eligibility (cradValidTitleApprovalWhereSql): status, coordinator_status, and crad_status all Approved; all three signature fields non-empty; active coordinator + adviser assignments. Codes: TAP-? / RG-YYYY-NNN.



1 Title submission

Student Portal research-proposal-submission.php (CRAD FORM S2 V3) ? api/send-to-adviser.php inserts title_approvals (Pending; coordinator/crad Not Ready).

2 Triple approval

Adviser Approved/Returned ? Coordinator screening ? CRAD Officer register-proposal.php (rpTitleApprovalApprove + crad_signature_data). Returned notifies student.

3 Research Group Number

research-group-number.php creates research_groups linked to title_approval_id; then research-coordinator-management assigns Active coordinator.

4 Adviser & Panel Assignment

Coordinator allowlisted pages assign research_adviser_assignments / research_panel_assignments; panel notifications cleaned when registry invalid.

5 Chapter Submissions

submit-chapters.php ? chapter_submissions; grammarian scores chapter_evaluations (Submitted ? Under Review ? Needs Revision ? Accepted).

6 Progress Monitoring

research_plans / research_milestones / research_progress_updates + AI analyses; faculty monitoring pages.

7?10 Defense path

research-defense-scheduling; preoral/final evaluations; research_revision_cycles; manuscript_* ? final_manuscript_approvals.

11?14 Outputs

publications, document-repository, research-analytics-reporting, grant dashboards.

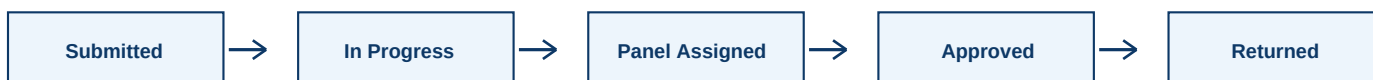
4. Capstone Workflow (Detailed)

4.1 Primary state machine ? title_approvals



Any tier may Return (with signature/notes). Only full triple Approved + signatures unlocks official registry operations (chapters eligibility, panel notifications, etc.).

4.1b Legacy ? research_proposals (still in schema)



Older CRD-YYYY-NNNN tracking with proposal_members / proposal_documents / proposal_status_logs. proposal-submission-tracking.php currently redirects toward register-proposal ? dual-path drift to be aware of.

4.2 Chapter evaluation loop



Tables: chapter_submissions, chapter_submission_history, chapter_evaluations, chapter_evaluation_notifications. Faculty evaluation-scoring and CRAD reviewer-evaluation pages participate.

4.3 Defense path



Tables: research_defense_schedules, research_venues, preoral_defense_evaluations, final_defense_evaluations, final_defense_recommendations, manuscript_submissions, manuscript_evaluations, final_manuscript_approvals.

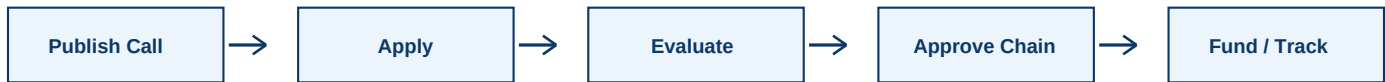
4.4 Notification spine

Shared includes/notifications.php + CRAD chapter/panel/progress/grant notification tables. Events (assignment, under review, needs revision, accepted, final MS approved) create in-app notifications with deep links back to the correct portal page.

5. Research Grant & Funding Workflow

Grant management is a first-class CRAD sub-system (module key `crad_grant`). Live APIs poll with fingerprints for near-real-time UI updates (`assets/js/grant-*-live.js`).

5.1 Opportunity ? Funding pipeline



Sequential approval chain (enforced in `grant-approval-helpers.php`)



Workflow tables: `grant_proposal_approval_workflows` + `grant_proposal_approval_steps`. Statuses per step: Queued / Pending / Approved / Rejected. Monitor UI: `approval-workflows.php` and live JS `crad-approval-dashboard-live.js`.

5.2 Post-funding

- ? budget-disbursement / `grant_funding_disbursements` ? release of funds.
- ? funded-research + `grant_funded_progress_evidence` ? progress evidence uploads.
- ? project-milestones / `grant_funded_project_milestones` ? milestone tracking.
- ? publications-ip / `grant_publications_ip_repository` ? outputs & IP.
- ? document-repository ? centralized grant documents.
- ? core-system-dashboard & dashboard-analytics ? live KPI cards from DB (no mock data).

6. Key Screens, APIs & Data

6.1 Representative CRAD pages

- ? index.php ? module overview grid
- ? register-proposal / proposal-submission-tracking / revise-proposal
- ? research-group-number / capstone-group-student-registry
- ? adviser-panel-assignment / assign-research-adviser / assign-panel-members
- ? research-defense-scheduling / panel-availability / retrieve-defense-ready-research
- ? final-manuscript-review / final-manuscript-approval
- ? grant-opportunities / proposals-applications / approval-workflows / budget-disbursement
- ? funded-research / project-milestones / publications-ip / document-repository
- ? research-analytics-reporting / core-system-dashboard / research-repository
- ? send-notifications / research-collaboration-portal

6.2 JSON APIs (modules/crad/api/)

- ? research-progress.php, adviser-progress.php, chapter-live.php, chapter-document.php
- ? ai-document-analysis.php ? adviser-only AI review of progress attachments
- ? grant-management.php, grant-approval.php, grant-evaluation.php, grant-funding.php
- ? grant-milestones.php, grant-funded-research.php, grant-document-repository.php, grant-final-output.php
- ? send-to-adviser.php, final-manuscript-document.php, progress-document.php

6.3 Core tables in crad_db (selected)

- ? HUB: title_approvals (status + coordinator_status + crad_status + 3 signatures)
- ? research_groups ? title_approval_id; research_coordinator_assignments
- ? research_adviser_assignments, research_panel_assignments, panel_* notifications
- ? Legacy: research_proposals, proposal_members, proposal_documents, proposal_status_logs
- ? chapter_submissions, chapter_evaluations, manuscript_*, final_manuscript_approvals
- ? research_plans, research_milestones, research_progress_*, research_progress_ai_analyses
- ? research_defense_schedules, research_venues, *defense_evaluations, research_revision_cycles
- ? grant_opportunities, grant_applications, grant_proposal_*, grant_funding_disbursements
- ? grant_funded_*, grant_document_repository*, grant_publications_ip_repository, publications

7. Architecture & Integration Map

Presentation: PHP pages + Bootstrap UI + live JS pollers

Application: crad/includes helpers, APIs, auth gates, notifications

Domain DBs: crad_db (research) + sms2_db (users, roles, activity_logs)

Cross-module: Student Portal | Faculty | Payment | User Mgmt | Reports

Integrations

- ? Student Portal starts proposals, chapters, progress, final manuscript.
- ? Faculty workspace hosts adviser monitoring, defense schedule, evaluation scoring.
- ? User Management owns RBAC (role_permissions) and account lifecycle.
- ? Account Security: password policy, OTP/Authenticator (TOTP), logout, passkeys.
- ? Audit: activity_logs via logActivity() with user, role, IP, user-agent, timestamp.
- ? Crypto: AES-256-GCM (includes/crypto.php) for secrets/TOTP and encrypted backups.

CRUD pattern in CRAD

Most CRAD desks follow a list/process UI (includes/crad-module-process.php ? module-process-list-view.php) with actions: new, validate, approve, report, view, edit, archive, restore, delete/submit. Grant and progress areas use richer form+API CRUD with file uploads (PDF/DOCX) and status transitions rather than simple row delete.

8. Verification Checklist Mapping

Legend: PRESENT = implemented and evidenced in code; PARTIAL = present with gaps or limited scope; ABSENT = not found / not applicable to this campus SMS deployment.

Section 1 ? Core IT Capabilities & Emerging Technology

PRESENT

End-to-End Workflow Logic

Full research + grant pipelines exist with DB-backed status transitions, validations, and notifications. Some catalog pages still show illustrative sample rows alongside live queries.

PRESENT

User Authentication Workflow

Login/logout, session auth (requireAuth), password reset requests, must_change_password, OTP/Authenticator confirmation, and module security screens are implemented.

PRESENT

CRUD Operations

Create/Read/Update across proposals, groups, assignments, grants, milestones, publications; archive/restore pattern for process lists; hard deletes restricted for users.

PRESENT

AI Integration

Adviser AI document analysis API (ai-document-analysis.php) stores results in research_progress_ai_analyses; role-gated to advisers.

ABSENT

IoT Integration

No IoT device ingest, MQTT, or sensor telemetry modules found. (IoT appears only as sample research titles, not system capability.)

PARTIAL

API Integration

Internal JSON REST-style endpoints under modules/crad/api/ with session auth and role checks. No public third-party OpenAPI gateway documented.

PARTIAL

Offline Synchronization

Offline-safe static assets (local vendors) exist; no offline transaction queue / sync engine for CRAD forms.

PARTIAL

Background Processing

Notification backfills and on-request jobs exist; no robust cron/scheduler service documented for CRAD reports.

PARTIAL

Error Recovery

Try/catch with error_log and user-facing DB error banners on key pages; no formal saga/compensating transaction framework.

PARTIAL

Scalability

MariaDB + PHP on XAMPP/typical LAMP; live pollers use fingerprints to reduce payload. No load-test evidence in repo.

Section 2 ? Security, Data Privacy & AI Governance

PRESENT

Multi-Factor Authentication (MFA)

Google Authenticator TOTP (includes/totp.php, authenticator-ui.php), email OTP for sensitive actions, optional WebAuthn passkeys (includes/passkey.php).

PRESENT

Role-Based Access Control (RBAC)

role_permissions matrix; smsRoleAllowedForModule(); role-specific sidebars; grant approval steps mapped to roles.

PRESENT

Password Security

password_hash(PASSWORD_DEFAULT); smsValidatePasswordStrength with configurable min length via system settings.

PRESENT

Account Lockout

failed_login_attempts + locked_until; configurable max_failed_logins / lockout_minutes; lockout logged to activity_logs.

PARTIAL

TLS Encryption

Application supports HTTPS deployments; local XAMPP default is HTTP. TLS 1.3 depends on hosting/Apache config, not app code.

PARTIAL

Database Encryption

AES-256-GCM used for secrets/TOTP and backup files (crypto.php). Full-database transparent encryption (TDE) not evidenced.

PARTIAL

Personal Data Protection

Role scoping limits PII visibility; security tips reference confidential research IP. Formal DPA compliance pack not in module.

ABSENT

Consent Management

No dedicated consent registry / privacy policy acceptance workflow found for research data subjects.

PARTIAL

Right to Delete Data

User accounts can be archived then permanently deleted by admin; no self-service data-subject erasure portal for research artifacts.

PRESENT

Audit Trail

activity_logs via logActivity(); proposal_status_logs; progress activity logs; grant workflow step history.

PARTIAL

AI Prompt Protection

AI endpoint is authenticated and adviser-scoped; dedicated prompt-injection defenses / content filters not clearly documented.

PARTIAL

Source Code Security

.htaccess present in module; auth on APIs. No committed SAST/DAST scan reports in repository.

PARTIAL

Dependency Security

Local vendor assets reduce CDN risk; no automated dependency CVE report checked into the project.

Section 3 ? Operational Analytics & Dashboards

PRESENT

Real-Time Dashboard

Grant/core dashboards and multiple *-live.js pollers refresh metrics/lists without full page reload using fingerprints.

PRESENT

Dashboard Accuracy

core-system-dashboard.php states counts are live SQL from grant_opportunities / grant_applications (no mock data).

PARTIAL

Interactive Charts

KPI cards and filtered lists are strong; advanced drill-down charting varies by page.

PRESENT

Historical Reports

Submission history, status logs, analytics reporting pages, and archived process records support historical review.

PRESENT

KPI Monitoring

Active studies, approval rates, defenses, publications, grant open/approved/denied counters.

PARTIAL

Report Export

Document/file downloads (PDF/DOCX) and curriculum export tool exist; universal PDF/Excel/CSV export for all CRAD KPIs is incomplete.

Section 4 ? Data Interoperability

PARTIAL

CSV Import

Limited/office-specific import patterns; not a universal CRAD bulk CSV importer.

PARTIAL

Excel Import

xlsx MIME supported for some file viewing/uploads; structured Excel import pipelines are limited.

PARTIAL

JSON Import

APIs accept JSON bodies for actions; batch JSON dataset import UI not a primary CRAD feature.

PRESENT

Invalid File Detection

Upload validators check MIME/extensions (e.g., PDF for proposals) and return API error messages.

PARTIAL

Bulk Upload

Multi-document uploads exist per record; large dataset bulk load tooling is limited.

PARTIAL

Export Accuracy

File serving preserves originals; formal export-format fidelity tests not packaged as evidence.

Section 5 ? Reporting System

PARTIAL

Custom Reports

research-analytics-reporting and reports-analytics module provide scoped reports; fully user-built report designer is limited.

PRESENT

Report Filters

Process lists support search/status/type filters; analytics pages filter by workflow state.

PARTIAL

Report Branding

UI branding exists; dedicated branded PDF report templates for all CRAD outputs are not consistently centralized.

ABSENT

Scheduled Reports

No evidenced automated email-scheduled CRAD report jobs.

PARTIAL

Print Functionality

Browser print works on HTML pages; dedicated print CSS varies by screen.

Section 6 ? Database Architecture

PRESENT

Database Normalization

Separate entity tables for proposals, members, documents, evaluations, grants, workflows ? normalized relational design.

PARTIAL

Foreign Key Integrity

Logical FKs widely used; some cross-DB links to sms2_db users are application-enforced rather than hard FKs.

PARTIAL

Data Dictionary

SQL comments on key columns + README install guide; formal published data dictionary document incomplete.

PRESENT

Index Optimization

Indexes on status, recipient, workflow keys appear in schema (e.g., panel/notification indexes).

PARTIAL

Query Performance

Fingerprint polling and LIMIT clauses help; no committed EXPLAIN/performance report.

PRESENT

Backup Procedures

System settings offer password-protected AES-256 backup download for admins.

PARTIAL

Restore Procedures

Backup tooling exists; documented restore drill evidence should be produced for audit.

Section 7 ? UI / UX & Accessibility

PRESENT

Responsive Layout

responsive.css + Bootstrap-based layouts used across CRAD/process list UIs.

PRESENT

Navigation

Role-aware sidebar groups (CRAD, Student Research, Faculty Monitoring, Grant Review).

PRESENT

Visual Consistency

Shared module-process-list CSS, banners, status badges, icon system.

PRESENT

Form Validation

Client + server validation on proposals, grants, password strength, upload types.

PARTIAL

Loading Indicators

Live JS areas update quietly; dedicated spinners not universal on every action.

PRESENT

Error Messages

User-facing alerts for DB/API failures and forbidden access.

PARTIAL

Keyboard Accessibility

Standard HTML controls are keyboard reachable; full WCAG keyboard audit not evidenced.

PARTIAL

Screen Reader Support

Some aria-labels/roles present (e.g., dashboards); comprehensive AT testing not evidenced.

PARTIAL

Color Contrast

Generally high-contrast admin UI; formal WCAG contrast evaluation report not in repo.

Checklist scoreboard (CRAD-relevant)

Approximate distribution across all checklist items above:

- ? PRESENT ? strong: authentication, RBAC, lockout, passwords, MFA, audit, CRUD, workflows, dashboards, KPIs, uploads validation, backups, responsive UI.
- ? PARTIAL ? improve for audit: TLS evidence, full DB encryption, privacy/consent docs, exports, scheduled reports, AI prompt hardening, accessibility reports, load tests.
- ? ABSENT / N/A ? IoT integration, consent management portal, scheduled report emails.

9. Gaps, Risks & Recommendations

Priority improvements for audit readiness

- ? 1. Produce Functional Test Report + Load Test Report for title-approval?defense and grant approval chains.
- ? 2. Enable HTTPS (TLS 1.2/1.3) in deployment and attach SSL Labs / server config evidence.
- ? 3. Publish a CRAD Data Dictionary + ER diagram derived from crad_db.sql (include title_approvals hub).
- ? 4. Resolve legacy vs modern dual path: retire or clearly label research_proposals vs title_approvals.
- ? 5. Fix README drift (still mentions obsolete submit-documents.php flow).
- ? 6. Add consent & data-subject request workflows if personal research participant data is stored.
- ? 7. Harden AI endpoint (prompt allow-list, output filtering, rate limits) and document AI governance.
- ? 8. Unify report export (PDF/Excel/CSV) with institutional header/footer branding.
- ? 9. Document backup restore drill results and retention policy; run SAST/dependency scans.
- ? 10. Replace catalog sample rows with live queries; align coordinator allowlist with visible nav.
- ? 11. Add WCAG keyboard/contrast spot-checks on top 10 CRAD screens.

Developer verdict

CRAD is a mature research-office system: students start titles in the Student Portal; Adviser ? Coordinator ? CRAD Officer approve with digital signatures; groups are registered; Faculty/Student drive progress and defense; a separate grant subsystem funds research through a six-step approval chain ? all in crad_db, gated by SMS2 roles. Checklist strength is high on workflows, RBAC, MFA, audit, and dashboards. Gaps are mainly IoT (N/A), privacy/consent packs, TLS/ops evidence, dual-path cleanup, and packaged test/scan reports.

End of Assessment

Source: sms2-capstone-main /modules/crad + shared auth/security includes